

LABORATORY RECORD

Course: Full Stack Web Development

Course Code:23CM4121

Experiment No.: 2

Student Name: G.Kavya

Roll No.: A24126552082

Date: 31-08-2026

TITLE

Design a Responsive College Webpage Using HTML5 and CSS3 Layout Cards

AIM

To understand and implement structural HTML5 elements combined with CSS3 styling rules (including Flexbox layout and hover animations) to build a stylized static webpage for a college website.

OBJECTIVES

- Create a structured HTML5 document with header, navigation bar, content sections, and footer.
 - Apply CSS styling for background colors, font styling, padding, margins, and text alignment.
 - Implement responsive layout cards using CSS Flexbox for displaying course options, facilities, and activities.
 - Incorporate interactive pseudo-classes (:hover) for navigation links and card scale transitions.
 - Utilize modern box sizing (box-sizing: border-box) and reset default margins.
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TASK DESCRIPTION

Design a visually styled webpage titled "My College Website". The layout consists of a centered dark header section, a navigation bar with interactive links, an "About Us" section containing descriptive text, and a three-card layout displaying details on "Courses", "Facilities", and "Activities". The cards must be styled with shadows, rounded borders, white background, and hover scaling effects. Finally, include a centered footer displaying copyright information.

ALGORITHM/PROCEDURE

1. Create an HTML document containing standard structural tags: html, head, header, nav, section, and footer.
 2. Define an external or embedded stylesheet to apply global reset parameters (margin: 0, box-sizing: border-box).
 3. Style the header and footer with dark backgrounds (#333), centered text, and white font colors.
 4. Format navigation links with horizontal spacing and a yellow hover color effect.
 5. Build a container class .cards using Flexbox layout rules to align three card modules horizontally with consistent spacing.
 6. Apply visual polish to individual .card blocks using background colors, borders, drop shadows, and scale transformations on mouse hover.
 7. Save and open the HTML file in a web browser to verify proper visual execution.
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PROGRAM / SOURCE CODE

The following code represents the complete HTML source code and CSS styling definitions:

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>My College Website</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

<header>
  <h1>My College Website</h1>
  <p>Welcome to our webpage</p>
</header>

<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
  <a href="#">Courses</a>
  <a href="#">Contact</a>
</nav>

<section>
  <h2>About Us</h2>
  <p>Our college provides quality education and helps students develop technical and professional skills.
  </p>

  <div class="cards">

    <div class="card">
      <h3>Courses</h3>
      <p>We offer various technical and professional courses.</p>
    </div>

    <div class="card">
```

```
<div class="cards">
  <div class="card">
    <p>Modern labs, library and other facilities are provided.</p>
  </div>

  <div class="card">
    <h3>Activities</h3>
    <p>Students can participate in sports and cultural activities.</p>
  </div>
</div>
</section>

<footer>
  <p>© 2026 My College Website</p>
</footer>

</body>
</html>
```

Stylesheet.css

```
* {
  box-sizing: border-box;
}

body {
  margin: 0;
  font-family: Arial, sans-serif;
  background-color: #f2f2f2;
}

header {
  text-align: center;
  background-color: #333;
  color: white;
  padding: 25px;
}

nav {
  text-align: center;
  background-color: #555;
  padding: 15px;
}

nav a {
  color: white;
  text-decoration: none;
  margin: 20px;
}

nav a:hover {
  color: yellow;
}
```

```

section {
  padding: 30px;
  text-align: center;
}

.cards {
  display: flex;
  justify-content: center;
  gap: 20px;
  margin-top: 25px;
}

.card {
  background-color: #ffffff;
  width: 250px;
  padding: 20px;
  border-radius: 10px;
  box-shadow: 0 2px 8px #888888;
}

.card:hover {
  transform: scale(1.05);
}

footer {
  text-align: center;
  background-color: #333333;
  color: #ffffff;
  padding: 15px;
  margin-top: 30px;
}

```

CODE EXPLANATION

HTML/CSS Element	Purpose & Explanation
header & h1	Defines the top banner section of the college webpage formatted with a dark background (#333) and centered white text.
nav & a	Creates navigation menu links ("Home", "About", "Courses", "Contact") with custom padding and hover effects.
nav a:hover	Applies interactive hover feedback changing text color to yellow when hovered over by the mouse cursor.
section & h2	Wraps the main body section containing introductory content and headline text.
.cards	Container element utilizing standard card structures arranged side-by-side with proportional gaps.

.card	Individual card container with white background, rounded corners (10px border radius), and visual depth provided by drop shadows.
.card:hover	Enhances user experience by applying dynamic scaling (transform: scale(1.05)) when hovering over individual card elements.
footer	Displays copyright notice aligned to center at the bottom of the page.

TEST CASES AND EXECUTION DATA

Test Case	Verification	Expected Result	Observed Result	Status
TC-01	Page header styling	Centered text on dark grey header banner	Rendered cleanly with correct contrast	Pass
TC-02	Navigation links & hover	White text links change to yellow on hover	Navigation bar renders correctly	Pass
TC-03	Card layout arrangement	Three cards aligned horizontally with gaps	Cards render side-by-side with padding	Pass
TC-04	Card box-shadow & radius	Soft drop shadow with rounded corners	Cards display rounded border and shadow	Pass
TC-05	Footer rendering	Centered copyright text in dark footer block	Footer stays aligned at page bottom	Pass

OUTPUT

The following rendering depicts the browser output representation for the executed HTML & CSS source code:

My College Website

Welcome to our webpage

[Home](#) [About](#) [Courses](#) [Contact](#)

About Us

Our college provides quality education and helps students develop technical and professional skills.

Courses

We offer various technical and professional courses.

Facilities

Modern labs, library and other facilities are provided.

Activities

Students can participate in sports and cultural activities.

© 2026 My College Website

RESULT

Thus, a styled static college webpage was successfully designed using HTML structural tags and CSS layout styles. The webpage demonstrates standard layout structure, flex card containers, box shadow styling, interactive hover transitions, and clean typography. The browser output screenshot confirms proper implementation of all design specifications.